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GROUP 16 - MANUFACTURING



A DISCRETE-EVENT SIMULATION STUDY OF THE FERMACORE TABLET LINE

SIMULATION IN OPERATIONS MANAGEMENT

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1 Introduction and Objective

FermaCore Labs operates a tablet production line combining batch-type and rate-based operations. Management's 2030 plan aims to maximize output within the existing capacity before committing to long-lead-time capital expansions.

The objective of this study is to use discrete-event simulation (DES) in AnyLogic to map the current system dynamics and identify operational bottlenecks. Using this baseline, we quantify the effects of targeted improvements (equipment additions and batch resequencing) and stress-test the line under rising 2030 demand (a growing order book) to locate the constraint that caps its output.

Scope

This report covers the mandatory simulation requirements: evaluating the current baseline, finding the bottleneck, and stress-testing the line under rising 2030 demand (a growing order book). We also include one optional analysis on Industry 4.0, specifically predictive maintenance at the tablet press, where sensors reduce scrap by catching press drift before it goes out of specification.

2 Statement of Collaboration

We did not collaborate with any other group on this project.

The work was carried out jointly throughout. Both authors developed the conceptual model and KPI set together and co-wrote the report. JACOB BRAMS took lead responsibility for the input modelling, output analysis and experimental design; GUSTAV MAGLEHØJ HANSEN took lead responsibility for the AnyLogic implementation and verification.

3 Performance Management

To evaluate the FermaCore line and to compare improvement scenarios, we track five Key Performance Indicators (KPIs), described with the S.M.A.R.T. framework against management's objectives. Four are primary and cover output (throughput), the bottleneck (utilisation), congestion (flow time and WIP) and schedule adherence (on-time delivery); one is secondary and covers material efficiency (yield). Table 1 gives the baseline value of each KPI, with the full results in Sections 8 and 9.

Key Performance Indicator	S.M.A.R.T.
Weekly throughput Baseline: 60 494.00 bottles	S: Bottles shipped from the packaging lines. M: Counted at the sink via bottlesShipped . A: Baseline sets the ceiling; improvements are compared to it. R: The output management wants to raise before any expansion. T: Per terminating 50-batch run.
Station utilisation Baseline: dryer 94.30 % (binding)	S: Busy share of drying, cleaning team and packaging. M: Busy hours over available hours, from the resource pools. A: Read directly from AnyLogic each run. R: Locates the bottleneck and shows if a fix only shifts it. T: Tracked across the whole run.
Flow time and WIP Baseline: 179.40 h / 19.4 batches	S: Time a batch spends in the line, and batches in process. M: meanFlowTime and avgWIP , linked by Little's Law. A: Logged from agent entry and exit times. R: Measures congestion and lead time as load grows toward 2030. T: Per batch, and time-averaged over the run.
On-time delivery Baseline: 98.00 % on time	S: Share of batches shipped on or before their due date. M: onTimePct and maxTardiness : completion vs. due date (planned + 332.00 h lead + 24.00 h grace; the 332.00 h lead is the baseline maximum flow time, so baseline is on time by construction and any lateness reflects volume growth). A: Evaluated at the ship event, not at release, so genuine lateness is captured. R: The case's one-day release constraint; tested under 2030 demand. T: Per batch, summarised per run.
System yield Secondary, baseline scrap 6.20 %	S: Good tablets kept versus tablets scrapped. M: scrapRate from premix, press and packaging losses. A: Computed from scrap events at the end of each run. R: The lever behind Industry 4.0 (press scrap 10.00 % to 4.00 %). T: Cumulative per terminating run.

Table 1: Performance metrics, S.M.A.R.T. justification and baseline value.

4 Modelling

One process-batch flows as a single agent through six operations, becoming a specific product only at coating. The plan gives each batch as 150.00 kg of active-ingredient powder blend; 75.00 kg of excipients is added in premix, giving a 225.00 kg batch of 300 000.00 tablets. Figure 1 is the high-level flow chart; Figure 2 is the activity diagram in Rossetti notation [1], laid out as a left-to-right flow that wraps across rows. Circles are queues, rectangles are activities with their delay distribution, and small circles are resources. Each resource use is marked with a striped half-moon for the seize and a solid half-moon for the release. Premix is drawn as its two physical units (old and new), which differ in speed and failure rate; identical parallel resources (presses, packaging lines) are kept as one block with a capacity count.

4.1 System flow chart

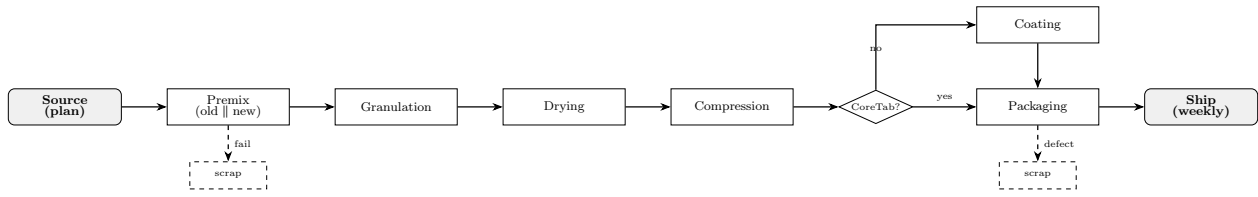


Figure 1: Simple flow chart of the FermaCore line. CoreTab skips coating; a premix failure or pre-packaging inspection scraps material.

4.2 Activity diagram

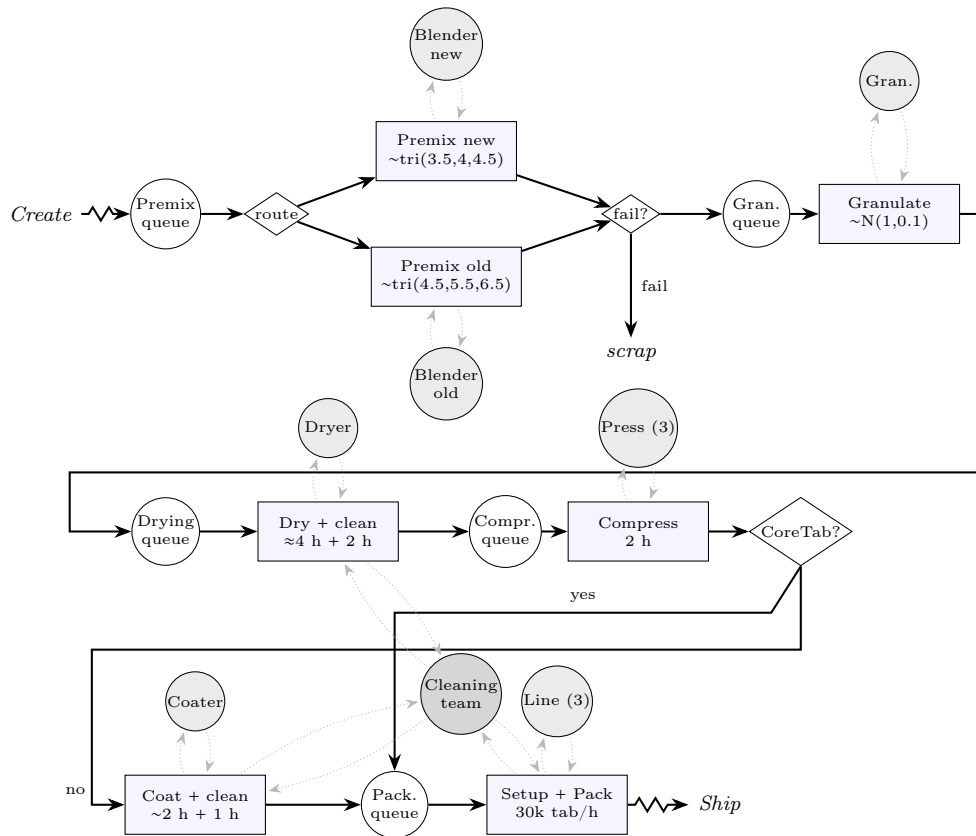


Figure 2: Activity diagram in Rossetti notation, read left-to-right and wrapping across three rows. Queues are circles, activities rectangles with their delay distribution, resources small circles. Each resource connects to its activity by two curved dotted lines: the seize (resource to activity) and the release (activity back to resource). Premix is drawn as its old and new units; the shared cleaning team is seized and released at the drying, coating and packaging cleans.

5 Documented Assumptions

The model represents one process-batch as a single agent. The production plan gives each batch as 150.00 kg of pre-weighed active-ingredient powder blend, and 75.00 kg of excipients is added in premix, so a batch weighs 225.00 kg. With 500.00 mg active and 250.00 mg excipient in each 750.00 mg tablet, one batch gives 300 000.00 tablets before yield losses ($\text{tablets} = \text{kg}_{\text{active}} \times 2000$). Batches never mix, because of full traceability, so each batch stays a separate entity from start to end.

The time unit is hours and the horizon is the five-day plan. The run is terminating and stops when the

last batch is shipped. Arrivals follow the **Plan** table exactly: 50 batches over 1–5 July, ten per day, with a mix of 17 CoreTab, 18 PlusTab and 15 MaxTab.

For routing and resources we make the following assumptions. Premix sends each batch to whichever unit, old or new, is free, and a failure during processing scraps the whole batch. Premix failure is represented solely through this probabilistic batch-scrap; the resource-pool breakdown/repair mechanism is configured but records zero repair events in every run, so the fitted repair distributions (§6) are not exercised in the baseline and do not affect any result. CoreTab skips coating, while PlusTab and MaxTab are coated. Packaging is rate-based at 30 000.00 tablets/hour per line, and leftover tablets that cannot fill a full bottle are scrapped. One shared **cleaningTeam** of capacity one performs all cleaning, setup, changeover and repairs through a seize-delay-release logic, which is why these activities compete for the same resource.

The yield assumptions are premix Triangular(98.5%, 99%, 99.5%), a baseline press scrap of 10.00 %, packaging Triangular(99%, 99.5%, 99.8%) and a drying yield of 100.00 %. Repairs and the cleaning team start empty and idle in each simulated week. The dryer queue uses FIFO discipline (**queueStrategy** = 0) in the baseline; the EDD and SPT alternatives are tested in Section 9 and leave throughput unchanged, because the drying times are near-homogeneous ($\sigma \approx 0.39$ h).

6 Input Modelling

Deterministic operation times and frequencies are taken directly from **FermaCore_input_data.xlsx**. The case requires three quantities to be estimated from the supplied observation sets with the distribution fitter: the *drying processing time*, the *premix/blending repair duration* and the *packaging-line repair duration*. Each was fitted and checked with a Kolmogorov–Smirnov (KS) goodness-of-fit test (Table 2, Fig. 3).

Quantity	Distribution	Fitted parameters	AnyLogic call
Drying time (h/batch), $n=300$	Normal	mean 4.00, std 0.39 (KS p = 0.92)	normal(0.39, 4.0)
Premix repair (h), $n=500$	Lognormal	mean 0.80, std 2.09 (KS p = 0.24)	lognormal(0.803, 2.086)
Packaging repair (h), $n=1000$	Lognormal	mean 5.01, std 7.67 (KS p = 0.66)	lognormal(5.009, 7.671)

Table 2: Fitted distributions, with the Kolmogorov–Smirnov goodness-of-fit p -value and the exact AnyLogic call (repair times in hours).

Drying-time fit. The drying observations are symmetric about a mean of 4.00 h (skewness ≈ 0). The model implements a **Normal(4.00, 0.39)** distribution, which gives the closest fit to the symmetric shape (Fig. 3, KS p = 0.92) and clearly outperforms a triangular over the observed range [2.703, 5.541] h. Because drying is the bottleneck, throughput is driven by the *mean* (4.00 h); the fitted mean lies more than ten standard deviations above zero, so the rare possibility of a non-positive Normal sample is immaterial in practice.

Given distributions (not fitted). Premix new Triangular(3.5, 4.0, 4.5) h; premix old Triangular(4.5, 5.5, 6.5) h; granulation Normal($\mu = 1$, $\sigma = 0.1$) h; compression 2.00 h/batch; coating Triangular(1.5, 2.0, 2.5) h; packaging 30 000.00 tablets/hour. Failure inter-arrival: packaging Exponential(mean 4 h); premix old Weibull($\beta=2.5$, $\alpha=1.5$) days; premix new Weibull($\beta=3.5$, $\alpha=0.5$) days. The two repair distributions in Table 2 were fitted as the case requires, but the resource-pool breakdown mechanism records zero events in the runs, so they are documented for completeness and do not influence the reported results; premix failure enters the model only through the probabilistic batch-scrap.

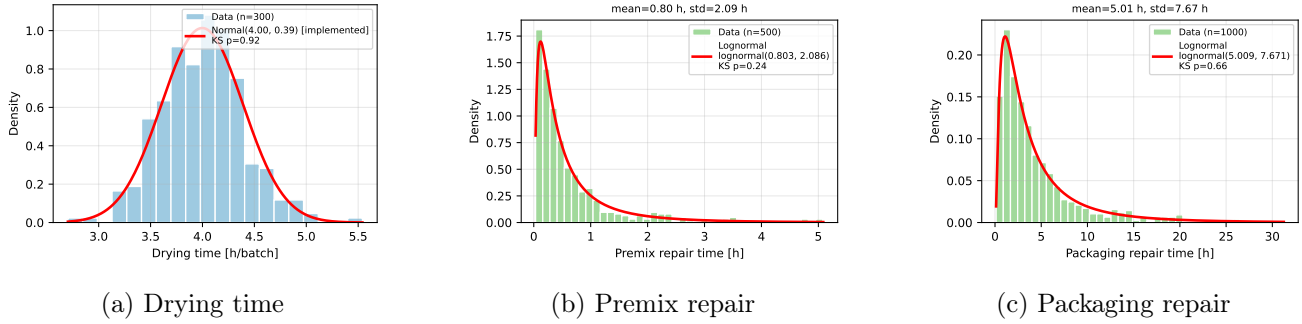


Figure 3: Distribution-fitter output for the three fitted quantities: empirical histogram with the fitted density overlaid (KS p -values in Table 2). For drying, the implemented Normal fit (KS $p = 0.92$) is overlaid on the histogram.

7 Verification and Validation

We verify and validate the model along the lines set out by Rossetti [1]: first confirming it was built right, then that it is the right model of the line.

7.1 Verification (building the model right)

We checked the implementation in four ways. First, we traced a single batch through every block to confirm the routing, the coating bypass for CoreTab, and the seize and release of the `cleaningTeam`. Two degenerate runs then gave the expected behaviour: with zero failures there is no repair scrap, and with one product there are no changeovers. A conservation check confirms that the tablets entering equal the tablets bottled plus the tablets scrapped, and a mass balance confirms that bottles times bottle size never exceed 300 000.00 per batch.

7.2 Validation and calibration (building the right model)

For validation we compare the model against analytic expectations. Drying carries the largest required machine-hours over the plan (about 300.00 h, Table 4), so it should be the binding constraint, ahead of the cleaning team and packaging. The simulated baseline utilisation confirms this ordering: drying 94.30 %, cleaning team 79.60 % and packaging 52.40 %. The per-station mean processing times also reproduce the input distributions they were drawn from. The one free modelling choice, namely how leftover tablets that cannot complete a bottle are rounded down, was fixed so that the baseline yield and throughput stay internally consistent before we ran the experiments.

Check	Expected	Simulated (baseline)	OK
Batch conservation	$\text{in} = \text{shipped} + \text{scrapped} = 50$	$46.5 + 3.5 = 50.0$	✓
Primary constraint	drying (largest unit-h)	drying 94.30 % util	✓
Constraint ordering	$\overline{\text{drying}} > \overline{\text{clean}} > \overline{\text{packaging}}$	$94.3 > 79.6 > 52.4$	✓
Little's Law	$\overline{\text{WIP}} \approx \lambda \overline{W}$	19.25 vs. 19.43 (0.90 %)	✓

Table 3: Verification and validation checks against the $n = 20$ baseline.

7.3 Analytical capacity check (predict, then confirm)

Before trusting the simulation we predict the bottleneck from first principles: the required machine-hours each operation must perform over the 50-batch plan.

Operation	h/batch (incl. clean)	Units	Batches	Required unit-h
Premix (old new)	≈4.75	2	50	≈119
Granulation	1.0	1	50	50
Drying (+2 h clean)	≈6.0	1	50	≈ 300
Compression	2.0	3	50	33
Coating (+1 h clean)	3.0	1	33	99
Packaging	≈9–10	3	50	≈150–167
Cleaning team (shared)	-	1	-	≈ 170

Table 4: Required machine-hours over the plan (predicts the constraint).

The analysis predicts **drying** as the primary constraint, the **cleaning team** second, and packaging third. The 2-hour drying clean couples the top two, since it both blocks the dryer and is the largest single draw on the cleaning team. Section 9.3 confirms this against the simulated utilisation.

7.4 Little’s Law consistency check

As a further validation we verify Little’s Law on the baseline, $WIP = \lambda \cdot \bar{W}$, linking our three flow KPIs:

$$\overline{WIP} \approx \text{throughput rate} \times \bar{W}, \quad (1)$$

where $\bar{W}$ is mean flow time. The throughput rate is taken *independently* of the WIP and flow-time statistics, from the shipped-batch count over the makespan: $\lambda = 46.5/433.4 = 0.1073$ batches/h. With $\bar{W} = 179.40$ h this predicts $\lambda \bar{W} = 19.25$ batches, against a directly measured $\overline{WIP} = 19.43$ batches, a 0.90% discrepancy that is well within sampling error. The three flow KPIs are therefore mutually consistent, which confirms that the model conserves work.

8 Output Analysis

This is a **terminating** simulation: the system starts empty and idle and ends when the 50th batch ships, so no warm-up period is deleted. Each scenario is run as n independent replications with a unique random seed per replication, using a **Parameter Variation** experiment that writes one row of KPIs per replication to CSV.

Number of replications. We size n with the half-width method [1]. For a KPI with sample mean $\bar{x}$ and sample standard deviation s over n_0 replications, the number of replications needed for a half-width ε at confidence $1 - \alpha$ is

$$n \geq \left(\frac{t_{n_0-1, 1-\alpha/2} s}{\varepsilon} \right)^2, \quad (2)$$

with $\alpha = 0.05$ and a target relative precision $\varepsilon \leq 5.00\%$ of the mean. We size on mean flow time, the noisiest of the reported operational KPIs; weekly throughput has lower relative variability and is therefore not the binding KPI for sizing.

We ran $n = 20$ replications of the baseline. Table 5 reports the achieved half-widths and the n that Eq. (2) requires for each KPI.

KPI	mean $\bar{x}$	st. dev s	half-width	rel. h-w	required n
Mean flow time (h)	179.4	6.95	3.25	1.80 %	3
Completion / makespan (h)	433.4	13.58	6.36	1.50 %	2
Throughput (bottles)	60,494	1,897	888	1.50 %	2
Avg. WIP (batches)	19.4	0.67	0.32	1.60 %	2
Dryer utilisation	0.943	0.009	0.004	0.40 %	1
Scrap rate	0.070	0.029	0.013	19.00 %	5*

Table 5: Replication sizing from the $n = 20$ baseline. Required n targets a 5.00 % relative half-width at 95.00 % confidence.

*Scrap rate has a small mean, so a 5.00 % *relative* target is not meaningful; we instead size it on an *absolute* half-width of ± 0.03 , which requires $n = 5$. The achieved absolute half-width at $n = 20$ is ± 0.013 (i.e. ± 1.3 percentage points).

Chosen replication count. Every reported KPI needs at most five replications to reach the target precision, so $n = 20$ leaves a comfortable margin and is used for all scenarios. All means in Section 9 are reported with their 95.00 % confidence intervals on this basis.

Resolving differences between scenarios. For scenario comparisons the relevant precision is the confidence interval on the *difference* from the baseline, not on a single mean. At $n = 20$ this resolves the one large effect, the Industry 4.0 scrap reduction, while the effect of any single capacity change on throughput falls below the detectable threshold. As shown in Section 9, that non-detectability is itself a result: shipped volume is fixed by the 50-batch plan and per-batch yield, so adding one unit of capacity shifts the bottleneck and changes flow time and utilisation, but does not move throughput by a statistically meaningful amount. The only lever that raises shipped output is the scrap reduction, which acts on yield rather than capacity.

9 Experimental Study

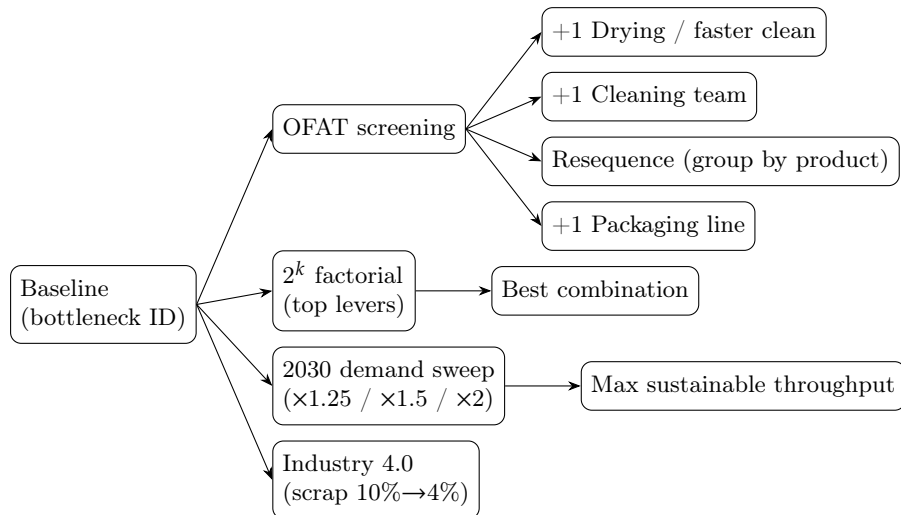


Figure 4: Scenario tree for the experimental study.

9.1 Scenario tree

We use a two-stage design: one-factor-at-a-time (OFAT) screening to rank the levers, followed by a 2^k factorial on the highest-impact levers, then a demand sweep for 2030 and the Industry 4.0 scrap scenario.

The OFAT branch is ordered to test the analytically predicted constraints (Sec. 7.3) first.

9.2 Factorial design

OFAT screening ranks the three capacity stations above resequencing, so the 2^3 factorial is run on drying, cleaning-team and packaging capacity. A full factorial (all eight corners, 20 replications each) is used rather than OFAT so that interaction effects are estimable: OFAT would miss them and can misread random variation between two runs as an effect.

Factor	Low (–)	High (+)
A: Drying capacity	1 dryer	2 dryers
B: Cleaning-team capacity	1	2
C: Packaging capacity	3 lines	4 lines

Table 6: 2^3 factor settings (low = –, high = +).

Each effect is the difference between the four high-corner means and the four low-corner means; its standard error is pooled from the within-corner variance across all eight corners ($SE = 347$ bottles, $df = 152$), so each effect carries a significance test.

Effect	Estimate (bottles)	t	Significant
C: packaging line	+454	1.31	no
AC	–277	–0.80	no
AB	–253	–0.73	no
A: dryer	–166	–0.48	no
B: cleaning team	+158	0.45	no
BC	–131	–0.38	no
ABC	–32	–0.09	no

Table 7: 2^3 effect estimates on weekly throughput (bottles). Sorted by magnitude. $|t| > 1.99$ is significant at 95.00 %.

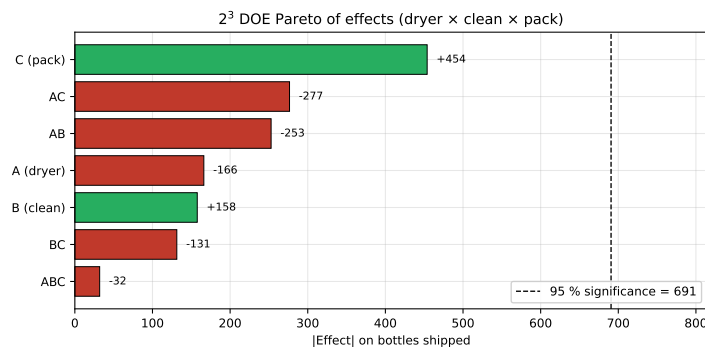


Figure 5: Pareto of 2^3 effects on throughput. No effect is significant; every estimate sits inside the noise band ($|t| < 2$).

Table 7 and the Pareto in Figure 5 show that *no effect is significant*: neither the three main effects (dryer, cleaning team, packaging) nor any interaction can be distinguished from zero, the largest being a +454-bottle packaging effect at $t = 1.31$. At the fixed 50-batch plan no capacity lever and no combination changes *shipped output*, a result the single-lever OFAT screening would have hidden, because output

is fixed by batches $\times$ per-batch yield, not by station capacity. Adding capacity moves the bottleneck (Sec. 9.4) but not the throughput. This conclusion is specific to the current order book: Section 9.5 shows that once the batch count itself grows toward 2030, capacity changes do shift lead time and on-time delivery, though far less than the investment would suggest.

9.3 Results: throughput across scenarios

Figure 6 compares weekly throughput across the scenario set with 95.00 % confidence intervals. Almost every bar overlaps the baseline: the capacity and sequencing levers do not move shipped output. The one clear exception is Industry 4.0 (press scrap 10 $\rightarrow$ 4.00 %), which lifts throughput by 4,842 bottles (+8.00 %, 95% CI [3,543, 6,141]), the only intervention whose interval clears the baseline by a wide margin.

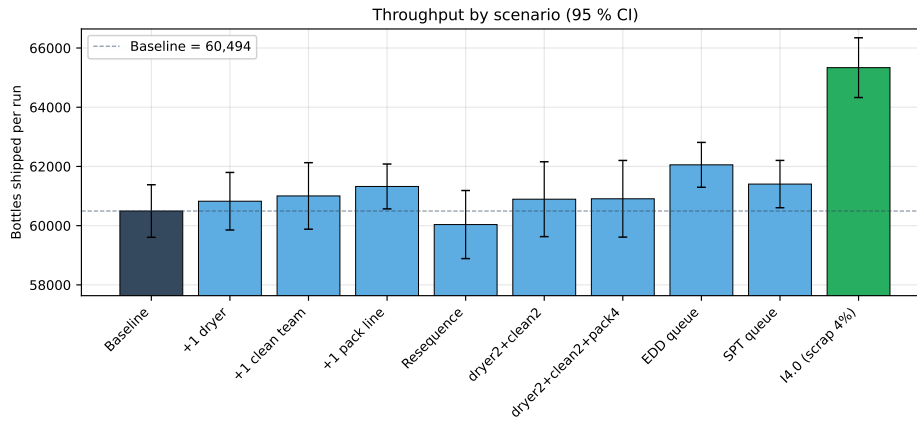


Figure 6: Weekly throughput by scenario with 95.00 % CI error bars. Industry 4.0 (green) is the only scenario that separates from the baseline.

Queue discipline. Switching the dryer queue from FIFO to EDD or SPT (`queueStrategy` = 1,2) leaves throughput within the plan-capped scatter (EDD +2.60 %, SPT +1.50 %), neither separated from the baseline. Drying times are near-homogeneous ($\sigma \approx 0.39$ h, Sec. 6), so reordering a queue of similar jobs cannot meaningfully change flow, and sequencing is not a lever here.

9.4 Bottleneck shift across scenarios

We report per-station utilisation, not only throughput, to show *which* resource binds as capacity is added. Table 8 and Figure 7 trace the bottleneck as it moves.

Scenario	Drying	Clean team	Packaging	Throughput
Baseline	94.3	79.6	52.4	60,494
+1 dryer	78.5	95.0	72.2	60,825
+1 dryer, +clean team	61.1	93.0	93.0	60,894
Best combination (+dryer, +clean, +pack)	78.2	92.8	76.7	60,908

Table 8: Mean utilisation (%) by station. Bold = binding constraint. The bottleneck moves from drying to the cleaning team, while throughput stays flat.

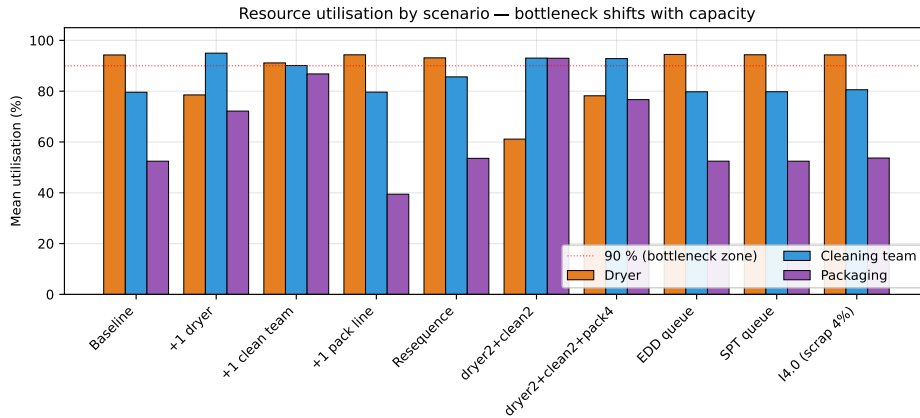


Figure 7: Resource utilisation by scenario. Adding a dryer pushes drying off the 90.00 % line but drives the cleaning team onto it, so the constraint shifts rather than clears.

Drying binds at baseline (94.30 %). A second dryer relieves it (78.50 %) but immediately loads the cleaning team to 95.00 %, because each dryer carries a 2-hour clean that draws on the same shared team. Adding a cleaning-team member then splits the load between the team and packaging, both near 93.00 %. Throughput never follows: it holds at $\approx 60,500$ – $61,000$ bottles across all four configurations. This is the bottleneck-shifting pattern: capacity relocates the constraint but cannot raise output, because output is plan- and yield-limited, not capacity-limited.

9.5 2030 demand and Industry 4.0

The 2030 question is one of *volume*: management expects demand to keep rising. We scale the order book through a `targetBatchCount` parameter that regenerates the plan at 50, 75 and 100 batches, holding the product mix (17/18/15) and the ten-batches-per-day cadence fixed, so only the volume changes. (A secondary `demandMultiplier` lens, which merely compresses the release of a fixed 50-batch plan, tests schedule resilience rather than the volume growth asked here.) Table 9 reports the baseline configuration at each volume.

Volume	Bottles	Dryer util.	Mean flow (h)	Avg WIP	On-time
$N = 50$ (today)	60,494	94.30 %	179	19.4	100.00 %
$N = 75$	92,649	96.20 %	260	28.3	71.00 %
$N = 100$	123,496	96.80 %	338	37.0	54.00 %

Table 9: Baseline configuration under genuine 2030 volume growth. Output scales with the order book, but lead time and on-time delivery degrade sharply as the single dryer saturates.

The line does *produce* the extra volume (60,494 to 123,496 bottles), but it does not absorb it gracefully. As the order book grows the single dryer saturates (94.30 % to 96.80 %), and the queue in front of it drives **mean flow time from 179 to 338 h**, average WIP from 19 to 37, and **on-time delivery from 100.00 % to 54.00 %** (worst-case tardiness reaches 296 h at $N = 100$). At current capacity the line meets higher 2030 demand only by letting lead times roughly double and service collapse (Figure 8).

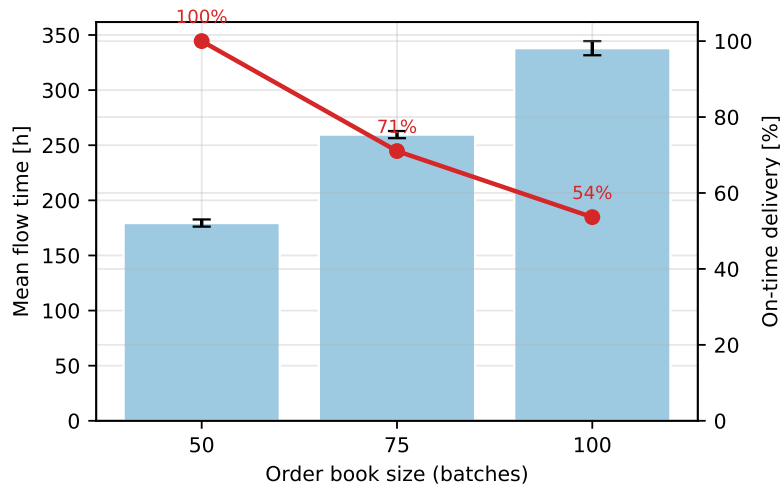


Figure 8: Service under 2030 volume growth: mean flow time (bars, 95.00 % CI) and on-time delivery (line) versus order-book size. Output scales with N , but lead time roughly doubles and on-time delivery falls from 100.00 % to 54.00 %.

Capacity combinations under volume growth. Adding capacity at volume shifts the constraint but barely recovers service. At $N = 100$ a second dryer relieves drying (96.80 % to 85.30 %) yet immediately saturates the shared cleaning team (97.50 %), because each dryer carries a 2-hour clean drawn from the same team; mean flow improves only 338 to 330 h and on-time delivery 54 to 55.00 %. Adding a cleaning-team member as well leaves the team still at 96.40 % and flow at 327 h. The dryer-clean coupling on the shared team is therefore a *structural* constraint that incremental capacity relocates but does not resolve. Restoring service at 2030 volume requires decoupling the dryer clean from the shared team (a dedicated or faster clean), not simply buying one more machine.

Industry 4.0: predictive maintenance at the press. The scenario models condition-based (predictive) maintenance: real-time sensors monitor press force and filling depth and flag the drift before the press leaves specification, so a developing fault is corrected before it scraps tablets rather than after. The QA estimate is that this cuts press scrap from 10.00 % to 4.00 %. Because output is bounded by batches $\times$ yield, raising good tablets per batch is the one intervention that lifts *shipped* throughput: 65,336 vs. 60,494 bottles, a gain of **4,842 bottles per run** (+8.00 %, 95% CI [3,543, 6,141]). It is the only lever that significantly raises shipped output, where no capacity lever does, and it needs only the sensor retrofit, not added line capacity. The same yield gain carries to volume: applied together with a second dryer at $N = 100$ (**dryer2_batches100**), the scrap reduction lifts good bottles from 122 448.00 to 130 345.00, a further 6.40 %.

9.6 Managerial value: benefit vs. effort

Lever	Δ throughput	Implementation effort / cost
Resequence (group by product)	-0.8% (n.s.)	Free: scheduling change only
+1 cleaning team	+0.3% (n.s.)	Low: additional shift / hire
+1 dryer	-0.3% (n.s.)	Medium: equipment, moderate lead time
+1 packaging line	+0.8% (n.s.)	High: capex, long lead time (the 2030 plan defers this)
Industry 4.0 (scrap 10 $\rightarrow$ 4.00 %)	+8.0%	Low: one-off press-sensor retrofit, no recurring cost

Table 10: Each lever's throughput gain against implementation effort. "n.s." = not statistically significant at 95.00 % (within the noise band).

Only one lever moves throughput at the current order book: the press-sensor retrofit (+8.00 %), at a fraction of the capital cost of any line investment. The capacity levers (more dryers, cleaning staff, packaging, resequencing) return nothing measurable on output at $N = 50$; their role appears only under volume growth (Sec. 9.5), where they shift the constraint and modestly ease lead time but do not, on their own, restore service.

10 Conclusions and Insights

The baseline identifies drying as the primary constraint, at 94.30 % utilisation and with the highest required unit-hours. The cleaning team is second at 79.60 %, and packaging, although it has the largest absolute workload, is only third at 52.40 % because it runs on three parallel lines. At the current 50-batch order book no capacity lever raises throughput: a second dryer relieves drying but moves the bottleneck onto the cleaning team (79.60 % to 95.00 %), since each dryer adds a two-hour clean, and adding cleaning staff then pushes the constraint onto packaging. Shipped output holds at 60 000.00 to 61 000.00 bottles in every case, because volume is fixed by the plan and per-batch yield, not by station capacity; the 2^3 factorial confirms that no effect, main or interaction, is significant.

Towards 2030 we grow the order book itself, to 75 and 100 batches. The line *produces* the higher volume (60,494 to 123,496 bottles) but does not absorb it: the single dryer saturates (94.30 % to 96.80 %), mean flow time roughly doubles (179 to 338 h), WIP nearly doubles, and on-time delivery collapses from 100.00 % to 54.00 %. Capacity additions at volume shift the constraint but barely recover service: a second dryer plus a larger cleaning team still leave the team near 96.00 % and flow at 327 h. The dryer-and-clean coupling on the shared team is a structural limit that more machines relocate rather than remove; restoring 2030 service needs the dryer clean decoupled from the shared team (a dedicated or faster clean), not just one more machine.

For shipped output the decisive lever is Industry 4.0. Predictive maintenance at the press, where sensors catch force or fill-depth drift before it scraps tablets, cuts scrap from 10.00 % to 4.00 % and raises throughput by 4,842 bottles per run (+8.00 %, 95% CI [3,543, 6,141]), the only lever that does so and the cheapest, needing only a sensor retrofit. Our recommendation is therefore twofold: at today's volume, defer the line capital expansion and invest in the press-sensor scrap reduction; and before 2030 volume materialises, address the dryer-clean coupling directly, since simply adding a dryer only moves the bottleneck to the cleaning team.

References

- [1] M. D. Rossetti, *Simulation Modeling and Arena*, 2nd ed. John Wiley & Sons, 2016.

Appendix

Software artefacts

AnyLogic model. `FermaCore_Replication_Harness_final.alp` is the model used for every result. It implements the full process flow (Sec. 4), the press scrap, a selectable dryer-queue rule (`queueStrategy`) and a configurable order-book size (`targetBatchCount`), and a Parameter-Variation experiment (`ReplicationsExp`) that runs $n = 20$ replications and writes one KPI row per replication to CSV. A scenario is run by setting the **Main** parameters (e.g. `numDryers`, `cleaningTeamCapacity`, `targetBatchCount`, `pressScrapRate`) and re-running the experiment.

Analysis scripts (Python).

- `halfwidth_analysis.py`: replication sizing (half-width method, required n per KPI) for one scenario CSV.
- `scenario_comparison.py`: cross-scenario KPI table with 95.00 % CIs, two-sample significance versus baseline, and the 2^3 factorial effects.
- `make_figures.py`: builds the throughput, utilisation, DOE-Pareto and volume figures from `data/`.
- `make_fit_figures.py`: builds the three input-distribution fit figures from `FermaCore_input_data.xlsx`.